Problem 3

a)

$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}^{\boldsymbol{T}}\boldsymbol{x}$, Where $A \in \mathbb{R}^{n\times n}$ is a symmetric matrix and $\boldsymbol{b} \in \mathbb{R}^n$

Wiring $f(x)$ using element-wise multiplication representation:

$$f(\boldsymbol{x}) = \frac{1}{2}\sum_{j}^{n}\sum_{i}^{n} x_i A_{ij} x_j + \sum_{k}^{n} b_k x_k \tag{1}$$

Partially differentiating $f(x)$ with respect to x_m:

$$\frac{\partial}{\partial x_m} f(\boldsymbol{x}) = \frac{1}{2}\sum_{j}^{n} A_{mj} x_j + \frac{1}{2}\sum_{i}^{n} A_{im} x_i + b_m \tag{2}$$

Projecting back to matrix representation:

$$\begin{bmatrix} \frac{\partial}{\partial x_1} f(x) \\ \vdots \\ \frac{\partial}{\partial x_n} f(x) \end{bmatrix} = \nabla f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{A}\boldsymbol{x} + \frac{1}{2}\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{x} + \boldsymbol{b} = \frac{1}{2}(\boldsymbol{A} + \boldsymbol{A}^{\boldsymbol{T}})\boldsymbol{x} + \boldsymbol{b} \tag{3}$$

Since A is symmetric and therefore $\boldsymbol{A} = \boldsymbol{A}^{\boldsymbol{T}}$, equation (3) can be rewritten as follows:

$$\nabla f(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b} \tag{4}$$

b)

$f(\boldsymbol{x}) = g(h(\boldsymbol{x}))$, Where g: $\mathbb{R} \to \mathbb{R}$ and f: $\mathbb{R}^2 \to \mathbb{R}$

$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial}{\partial_{x_1}} f(\boldsymbol{x}) \\ \vdots \\ \frac{\partial}{\partial_{x_n}} f(\boldsymbol{x}) \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial_h} g(h(\boldsymbol{x})) \times \frac{\partial}{\partial_{x_1}} h(\boldsymbol{x}) \\ \vdots \\ \frac{\partial}{\partial_h} g(h(\boldsymbol{x})) \times \frac{\partial}{\partial_{x_n}} h(\boldsymbol{x}) \end{bmatrix} = \begin{bmatrix} g'(h(\boldsymbol{x})) \times \frac{\partial}{\partial_{x_1}} h(\boldsymbol{x}) \\ \vdots \\ g'(h(\boldsymbol{x})) \times \frac{\partial}{\partial_{x_n}} h(\boldsymbol{x}) \end{bmatrix} \tag{5}$$

From equation (4), we get:

$$\nabla f(\boldsymbol{x}) = g'(h(\boldsymbol{x})) \times \nabla h(\boldsymbol{x}) \tag{6}$$

c)

$\nabla f(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}$, Where $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is a symmetric matrix and $\boldsymbol{b} \in \mathbb{R}^{n}$

$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} \left(\sum_{i}^{n} A_{1i}x_i\right) + b_1 \\ \vdots \\ \left(\sum_{i}^{n} A_{ni}x_i\right) + b_n \end{bmatrix} \tag{7}$$

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1}\left(\sum_{i}^{n} A_{1i}x_i\right) + b_1 & \cdots & \frac{\partial}{\partial x_1}\left(\sum_{i}^{n} A_{ni}x_i\right) + b_n \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_n}\left(\sum_{i}^{n} A_{1i}x_i\right) + b_1 & \cdots & \frac{\partial}{\partial x_n}\left(\sum_{i}^{n} A_{ni}x_i\right) + b_n \end{bmatrix} \tag{8}$$

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} A_{11} & \cdots & A_{n1} \\ \vdots & \ddots & \vdots \\ A_{1n} & \cdots & A_{nn} \end{bmatrix} = \boldsymbol{A}^{\boldsymbol{T}} = \boldsymbol{A} \tag{9}$$

Notice that $\boldsymbol{A}$ is symmetric and hence $\boldsymbol{A} = \boldsymbol{A}^{\boldsymbol{T}}$.

d)

$f(\boldsymbol{x}) = g(\boldsymbol{a}^{\boldsymbol{T}}\boldsymbol{x})$, Where g: $\mathbb{R} \rightarrow \mathbb{R}$ and $a \in \mathbb{R}^n$

Let $h(\boldsymbol{x}) = \boldsymbol{a}^{\boldsymbol{T}}\boldsymbol{x}$, hence $g\left(\boldsymbol{a}^{\boldsymbol{T}}\boldsymbol{x}\right) = g(h(\boldsymbol{x}))$

$$\nabla f = \frac{\partial}{\partial h} g(h(\boldsymbol{x})) \times \frac{\partial}{\partial x} h(\boldsymbol{x}) \tag{10}$$

Notice that $\frac{\partial}{\partial h} g(h(\boldsymbol{x}))$ is simply $g'(h(\boldsymbol{x}))$

Form equation (10)

$$\nabla f = g' \times \frac{\partial}{\partial x} h(\boldsymbol{x}) = g' \times \frac{\partial}{\partial x} \sum_{i}^{n} a_i x_i = g'(h(\boldsymbol{x}))\,\boldsymbol{a} \tag{11}$$

From equation (11):

$$\nabla^2 f = \begin{bmatrix} \frac{\partial}{\partial x_1} g'(h(\boldsymbol{x}))\,a_1 & \cdots & \frac{\partial}{\partial x_1} g'(h(\boldsymbol{x}))\,a_n \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_n} g'(h(\boldsymbol{x}))\,a_1 & \cdots & \frac{\partial}{\partial x_n} g'(h(\boldsymbol{x}))\,a_n \end{bmatrix} \tag{12}$$

From equation (12):

$$\nabla^2 f = \begin{bmatrix} g''(h(\boldsymbol{x}))\; a_1\, a_1 & \cdots & g''(h(\boldsymbol{x}))\; a_n\, a_1 \\ \vdots & \ddots & \vdots \\ g''(h(\boldsymbol{x}))\; a_1\, a_n & \cdots & g''(h(\boldsymbol{x}))\; a_n\, a_n \end{bmatrix} = g''(h(\boldsymbol{x})) \times \boldsymbol{a}\boldsymbol{a}^{\boldsymbol{T}} \tag{13}$$